

LOS ANGELES HAWTHORNE, CA, USA

WMO: 722956

Lat: 33.923N

Lon: 118.334W

Elev: 19

StdP: 101.09

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03167

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.9	7.7	-11.4	1.4	18.7	-8.3	1.9	17.8	8.0	14.8	6.9	15.3	0.5	70	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	31.4	17.5	29.2	17.6	27.5	17.7	21.5	26.7	20.8	25.7	20.2	25.0	3.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.4	24.2	18.9	13.7	23.7	18.2	13.1	22.8	62.5	26.7	60.0	26.0	58.0	25.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.3	5.6	4.1	36.2	1.6	2.8	3.0	38.2	2.1	39.8	1.2	41.4	0.0	43.4		
			0.8	22.4	1.2	0.9	-0.1	23.1	-0.8	23.6	-1.5	24.1	-2.3	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	14.9	14.8	15.9	16.8	18.1	19.6	21.9	21.8	20.2	17.5	14.5
		DBStd	3.68	2.84	2.78	2.61	2.65	2.12	1.67	2.03	1.95	2.52	2.94	2.42
		HDD10.0	1	1	0	0	0	0	0	0	0	0	0	1
		HDD18.3	577	112	105	84	62	28	4	1	1	11	48	121
		CDD10.0	2973	153	134	184	203	249	287	360	370	353	317	224
		CDD18.3	507	6	6	10	14	20	42	102	112	103	69	23
		CDH23.3	2229	64	57	78	109	101	83	313	368	456	397	173
		CDH26.7	588	10	14	21	37	37	16	49	62	133	151	56
(14)	Wind	WSAvg	2.1	1.4	1.9	2.2	2.6	2.7	2.6	2.4	2.2	1.8	1.4	1.3
(15)	Precipitation	PrecAvg	307	66	72	46	17	3	2	1	5	10	34	48
		PrecMax	749	323	338	162	115	24	25	9	32	48	96	190
		PrecMin	95	0	0	0	0	0	0	0	0	0	0	0
		PrecStd	141	72	75	42	23	5	5	2	4	11	16	39
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	27.9	28.9	29.1	31.5	31.4	28.9	30.4	30.6	33.9	34.5	31.7	26.5
		0.4% MCWB	12.9	13.7	14.5	15.3	15.6	19.4	21.2	20.7	20.1	16.9	15.2	12.3
		2% DB	25.0	24.7	25.4	26.2	25.9	25.4	27.8	28.4	30.1	30.6	27.9	23.4
		2% MCWB	11.7	12.3	13.1	13.9	16.1	18.6	20.5	20.2	19.6	16.1	14.2	11.8
		5% DB	22.6	21.8	22.4	23.0	22.8	23.7	26.3	26.8	27.7	27.6	24.8	21.2
		5% MCWB	11.6	12.3	13.0	13.9	16.2	18.0	20.1	19.9	19.5	16.4	13.2	11.4
		10% DB	20.0	19.0	20.1	20.9	21.2	22.6	25.0	25.3	25.8	24.8	22.3	18.9
		10% MCWB	11.4	12.3	13.2	13.8	15.6	17.4	19.6	19.5	19.1	16.7	13.4	11.5
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	15.8	16.1	16.7	17.1	19.0	20.2	22.4	22.0	22.4	21.0	17.7	16.2
		0.4% MCDB	19.3	20.5	23.2	24.4	26.1	26.5	28.2	27.4	29.3	26.2	23.0	18.5
		2% WB	14.9	15.0	15.7	15.8	17.6	19.1	21.3	21.2	21.4	19.7	16.8	15.2
		2% MCDB	18.0	18.8	20.7	21.9	22.7	24.2	26.2	26.3	26.9	25.3	21.3	18.1
		5% WB	14.2	14.3	14.9	15.2	16.8	18.6	20.6	20.6	20.5	18.7	16.1	14.4
		5% MCDB	17.7	18.0	19.3	20.4	21.3	23.1	25.1	25.4	25.7	23.6	20.3	17.3
		10% WB	13.5	13.6	14.3	14.6	16.0	17.9	20.0	19.9	19.8	17.9	15.4	13.6
		10% MCDB	17.1	17.4	18.0	19.1	20.1	21.7	24.2	24.6	24.7	22.3	19.6	16.7
(35)	Mean Daily Temperature Range	MDBR	9.6	8.6	7.8	7.4	6.2	5.8	6.3	6.5	7.1	8.1	9.3	9.3
		5% DB MCDBR	14.3	13.6	12.3	12.0	9.7	7.6	7.8	8.1	10.3	13.6	14.4	13.5
		5% WB MCWBR	6.1	5.9	5.1	4.8	3.8	3.2	3.0	3.2	3.9	5.2	6.4	6.7
		5% WB MCDBR	9.0	9.3	8.6	8.8	7.7	7.2	7.3	7.4	8.6	9.3	9.7	8.7
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.353	0.377	0.379	0.383	0.399	0.399	0.381	0.367	0.374	0.323	0.325
		taud	2.488	2.461	2.366	2.372	2.392	2.348	2.415	2.448	2.480	2.430	2.574	2.570
		Ebn at Noon	869	895	896	908	903	884	881	895	895	858	885	869
		Edh at Noon	86	98	117	121	120	125	117	111	103	100	79	76
(44)	All-Sky Solar Radiation	RadAvg	3.15	4.04	5.40	6.61	7.13	7.42	7.42	7.06	5.96	4.53	3.51	2.86
		RadStd	0.23	0.40	0.31	0.34	0.49	0.46	0.41	0.24	0.25	0.34	0.22	0.22

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.76	N/A	-1.19	+1.22	N/A	N/A	N/A	-163	+340	+177
(47)	Regional (52 neighbors)	+0.62	N/A	-1.64	+0.72	N/A	N/A	N/A	-116	+220	+105

Nomenclature: See separate page